Backend Schema & API Contracts

AstroIllustrator | Consolidated specification v2.1 | 22 October 2026

1. Storage policy

SQLite stores app configuration, job metadata, presets, illustration manifests and asset references. Large astronomical catalogs and generated SVG/PNG/PDF assets live as versioned files; avoid importing millions of catalog rows into SQLite without a query requirement. Use UUID IDs, UTC timestamps, foreign keys and schema migrations. Do not persist secrets or raw model credentials.

2. Core relational entities

Illustrations(Id, FigureCode UNIQUE, Title, IllustrationType, SchemaVersion, Language, PresetId, LatestJobId, CreatedAtUtc). RenderJobs(Id, IllustrationId, RequestJson, RequestHash, Status, RendererVersion, StartedAtUtc, CompletedAtUtc, ErrorCode, ErrorMessage). RenderOutputs(Id, JobId, Format, FilePath, Sha256, WidthPx, HeightPx, PrintWidthMm, PrintHeightMm). Presets(Id, Name UNIQUE, IllustrationFamily, SettingsJson, Version, IsBuiltIn). CatalogSnapshots(Id, Name, Version, Epoch, Frame, SourceUrl, License, Sha256, LocalPath). Assets(Id, AssetType, FilePath, Sha256, SourceOrProvider, ModelVersion, LicenseNotes, CreatedAtUtc). PromptInterpretations(Id, JobId NULL, PromptText NULL, Provider, Model, StructuredRequestJson, ValidationStatus, CreatedAtUtc).

3. Astronomical and template definitions

AsterismDefinitions(Code UNIQUE, DisplayName, Version, Source, License, StarIdsJson, EdgesJson) references stable catalog identifiers. ConstellationLineSets(Code, Version, Source, License, EdgesJson) remain versioned scientific resources. DiagramTemplates(Code UNIQUE, Version, ParameterSchemaJson, FormulaNotes, ApproximationPolicy, LabelKeysJson). Store observer/time/projection in immutable RenderJobs.RequestJson, not only a human-readable caption.

4. Typed request contract

IllustrationRequest: schemaVersion, illustrationType, figureCode, language, presetId, export, plus exactly one of skyChart, scientificDiagram or artisticComposite. Validate the discriminator and reject incompatible payloads. SkyChart payload: sceneType, constellationCodes[], asterismCode?, observer{latitude,longitude,elevationMeters}?, timeUtc?, projection, fieldOfViewDegrees, azimuthCenterDegrees?, magnitudeLimit, layers. ScientificDiagram payload: diagramType, mode (Schematic | Computed), observer?, timeUtc?, parameters{starId?, altitudeDegrees?, azimuthDegrees?, showZenith?, showHorizon?, showAngleArc?}. Use a typed per-template validator rather than unrestricted arbitrary JSON. ArtisticComposite payload: scientificRenderOutputId, backgroundAssetId, overlayPolicy=LockedGeometry.

5. Example request — Polaris

POST /api/illustrations/render with illustrationType=ScientificDiagram, scientificDiagram.diagramType=StarAltitude, mode=Computed, observer={latitude:42.6629,longitude:21.1655}, timeUtc=2026-07-15T21:00:00Z, parameters={starId:Polaris,showZenith:true,showAngleArc:true}, export={format:SVG}, language=sq. The renderer calculates and returns exact angle; the client must not hardcode 42.7 degrees.

6. Endpoints

GET /api/constellations; GET /api/asterisms; GET /api/diagram-templates; GET /api/presets; POST /api/illustrations/render; GET /api/render-jobs/{id}; GET /api/render-jobs/{id}/outputs; POST /api/ai/interpret (optional); POST /api/background-assets (optional). Return 400 for invalid input, 422 for scientifically unsupported combinations, 404 for missing definitions, 503 for unavailable optional AI provider. Include problem-details response and correlation ID.

7. Indexes and migrations

Unique: Illustrations.FigureCode, Presets.Name, AsterismDefinitions.Code. Index: RenderJobs(IllustrationId, CreatedAtUtc), RenderJobs(Status, CreatedAtUtc), RenderOutputs(JobId), CatalogSnapshots(Name, Version). Keep database-provider-specific features out of Core; EF Core migrations are in Infrastructure. PostgreSQL is a future deployment option, not a v2 requirement. Document control: This version supersedes earlier 1.0 and 1.1 scope descriptions. Keep requests, manifests, diagrams and AI integrations aligned with this specification.

v2.1 additions: global observer support, time-aware conjunctions and AstroIllustrator naming

New request discriminators and event DTOs

SkyChartRequest.sceneType adds AstronomicalEvent. EventRequest has eventType (PlanetaryConjunction | MoonPlanetConjunction | PlanetaryAlignment | TwilightVisibility), bodies[], eventSelection (FixedDate | DateRangeSearch), observer, timeReference, viewing, display and output. Reject mismatched combinations; do not rely on generic dictionary bags.

TimeReference and observer contracts

ObserverLocation {latitude, longitude, elevationMeters?, timeZoneId?}. TimeReference is either AbsoluteUtc {instantUtc}, LocalClock {localDateTime,timeZoneId,ambiguityPolicy}, or SolarRelative {localDate,anchor:Sunset|Sunrise|CivilDusk|CivilDawn|NauticalDusk|AstronomicalDusk,offsetMinutes}. Location validation is mandatory for horizon/event views.

Event results and provenance

CalculatedEvent {requestedDate, observationUtc, location, bodyPositions[{body, altitudeDeg, azimuthDeg, apparentMagnitude?}], angularSeparationDeg?, solarAltitudeDeg, twilightClass, visibleAboveGeometricHorizon, warnings[], ephemerisId, ephemerisHash, refractionMode, computationVersion}. Store calculated result alongside immutable request; never accept model-authored positions as authoritative.

Metadata and endpoints

Add ObserverPresets(Id,Name,Latitude,Longitude,ElevationMeters,TimeZoneId); EventSearchJobs(Id,RequestJson,Status,ResultJson,CreatedAtUtc); optional EventCandidates(JobId,InstantUtc,SeparationDeg,VisibilityFlags). Add GET /api/observer-presets; POST /api/astronomy/solar-events; POST /api/astronomy/events/search; POST /api/illustrations/render (event payload). Index EventSearchJobs(Status,CreatedAtUtc).

Example event request

{ "schemaVersion":"2.1", "illustrationType":"SkyChart", "skyChart": { "sceneType":"AstronomicalEvent", "event": { "eventType":"PlanetaryConjunction", "bodies":["Venus","Jupiter"], "observer":{"latitude":41.9028,"longitude":12.4964,"timeZoneId":"Europe/Rome"}, "timeReference":{"type":"SolarRelative","localDate":"2026-11-15","anchor":"Sunset","offsetMinutes":30}, "viewing":{"azimuthCenterDeg":270,"showHorizon":true,"showAngularSeparation":true}}}}. Illustrative request only: the date is not asserted to contain a conjunction.

Scientific note: illustrative dates and prompts are examples, not verified event predictions. All published charts require computed ephemerides and editorial review.